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1. Question 03

Part A

a. P_c = price consumer pay

P_p = price producer receive.

$P_c = P_p + 15$

$Q_d = 100 - 2P$

$= 100 - 2(P_p + 15)$

$= 100 - 2P_p - 30$

$= 70 - 2P_p$

$Q_s = 4P_p - 20$

$70 - 2P_p = 4P_p - 20$

$90 = 6P_p$

$P_p = 15$

$P_c = 15 + 15$

$= 30$

$\cancel{Q_s = 4(15) - 20}$ $\qquad$ $Q = 4(15) - 20$

$\cancel{Q_s = 40}$ $\qquad$ $Q = 40$

Price consumers pay $= 30$

price producers pay $= 15$

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b.

b.

$$Tax = 15 \times 40$$
$$= 600$$

price before tax = 20

consumars now pay - 30 (+10 burden)

producers now pay - 15 (-5 burden)

Total tax revenue = 600

consumers bear = $10 / unit

producers bear = $5 / unit

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Part B

a.

Before tax,

consumer surplus $= \frac{1}{2}(50-20)(60)$

$= \frac{1}{2}(30)(60)$

$= 900$

producer surplus $= \frac{1}{2}(20-5)(60)$

$= \frac{1}{2}(15)(60)$

$= 450$

Total welfare $= 900 + 450$

$= 1350$

After tax,

consumer surplus $= \frac{1}{2}(50-30)(40)$

$= \frac{1}{2}(20)(40)$

$= 400$

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producer surplus $= \frac{1}{2}(15-5)(40)$

$= \frac{1}{2}(10)(40)$

$= 200$

Tax revenue $= 600$

Total welfare $= 400 + 200 + 600$

$= 1200$

b.

DWL $= 1350 - 1200$

$= 150$

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Part C

a.

consumers are loss responsive to the change of the price when the demand is relatively inelastic. Because of that consumers bear a larger percentage of the tax burden.

b.